# A Framework for Step-by-Step Evaluation of Reasoning Chains in Large Language Models

鞠岩　JU YAN

京都大学情報学研究科 下平・本多研究室

2025/02

# Table of Contents


1. Introduction
2. Previous Methods
3. Proposed Method
4. Experiments
5. Conclusions

# Table of Contents


1. Introduction
2. Previous Methods
3. Proposed Method
4. Experiments
5. Conclusions

# Introduction

- Large Language Models (LLMs) have demonstrated remarkable reasoning capabilities through Chain-of-Thought (CoT) prompting, which guides the model to **break down problems into sequential reasoning steps**.

- **Key Challenges of Reasoning in LLMs:**
  - **Difficulty in Selecting Optimal Chains:** LLMs often generate multiple reasoning chains, but selecting the best chain is non-trivial.
  - **Error Propagation**: Mistakes in intermediate steps can accumulate, compromising the final output.
  - **Lack of Interpretability:** Existing methods fail to provide transparent evaluations of why certain reasoning chains are selected.

# Introduction

- **Why Addressing These Challenges is Critical:**
  - **Ensuring Reliability:** Robust evaluation mechanisms are necessary to identify logical and calculation errors in reasoning chains.
  - **Improving Interpretability:** Interpretability is essential to ensure trust and transparency in multi-step reasoning processes.
- **The Goal of This Thesis:**
  - Provide a method to **reliably identify the best reasoning chain** among multiple candidates.
  - Introduce detailed step-by-step evaluations to **enhance interpretability and error detection.**

# Table of Contents


1. Introduction
2. Previous Methods
3. Proposed Method
4. Experiments
5. Conclusions

# Previous Methods

- **Chain-of-Thought Prompting [1]**
  - Generates **intermediate reasoning steps** to improve LLMs' multi-step problem-solving ability.
  - **Limitation:** Lacks a mechanism for error detection

- **Complexity-Based Self-Consistency [2]**
  - Generates multiple reasoning chains, selects **the top-K most complex ones** based on reasoning steps, and determines the final answer through **majority voting.**
  - **Limitation:** Lack of Interpretability & Reliability

- **Self-Verification [3]**
  - Check each step of the reasoning process by using **backward verification.**
  - **Limitation:** High computational cost & Fails to specify root cause of problems.

# Table of Contents


1. Introduction
2. Previous Methods
3. Proposed Method
4. Experiments
5. Conclusions

# Proposed Method: Motivation & Idea

- **Limitations in Existing Methods:**

  ❌ **Lack of Fine-Grained Error Evaluation:** Existing methods struggle to distinguish between **logical errors** and **calculation errors**, making it hard to diagnose mistakes.

  ❌ **Complexity ≠ Correctness:** Selecting reasoning chains solely based on complexity may amplify **incorrect but lengthy** reasoning paths.

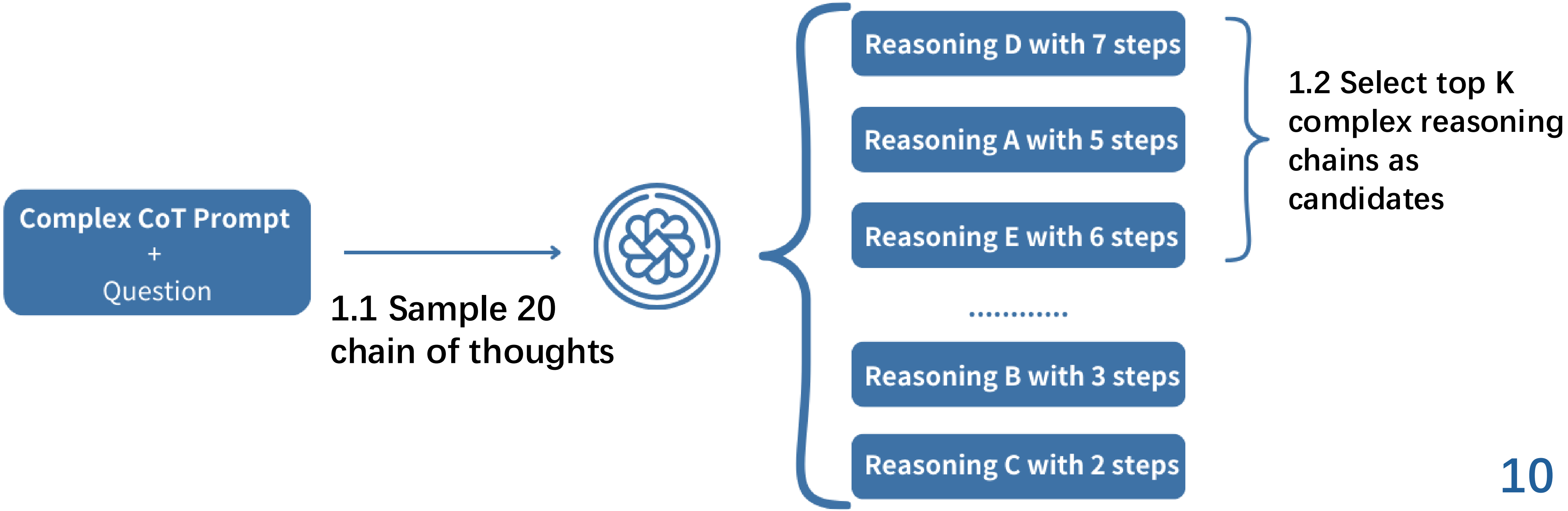
- **Our Solution:**

  ✅ Introduce **step-by-step error evaluation** to identify logical and calculation errors.

  ✅ Use a **scoring mechanism** to rank chains based on correctness, ensuring the **most reliable** reasoning chain is selected.

# Proposed Method

- **STEP 1: Generating and Selecting Candidate Chains**
  1. **Generate Diverse Reasoning Chains:** Use a **complex prompt** to sample 20 reasoning chains for a given question.
  2. **Select Top-$K$ Candidates:** Choose the $K$ **most complex chains** for further evaluation. (Complexity is based on the number of intermediate reasoning steps)



# Proposed Method

- **STEP 2: Constructing the Evaluation Prompt**
  - **Purpose:** Granular evaluation & Transparent scoring criteria
  - **Structure of the Evaluation Prompt:**

- **Grading rules:**
  - **Logic Accuracy (3 points)** → Assesses clarity, consistency, and reasoning soundness.
  - **Calculation Accuracy (2 points)** → Verifies mathematical correctness.

+

- **Scoring Mechanism:**
  - Each step is assigned a **score out of 5 points**.
  - **Total Score** = Sum of all step scores.
  - **Average Score per Step** = Total Score / Number of steps.

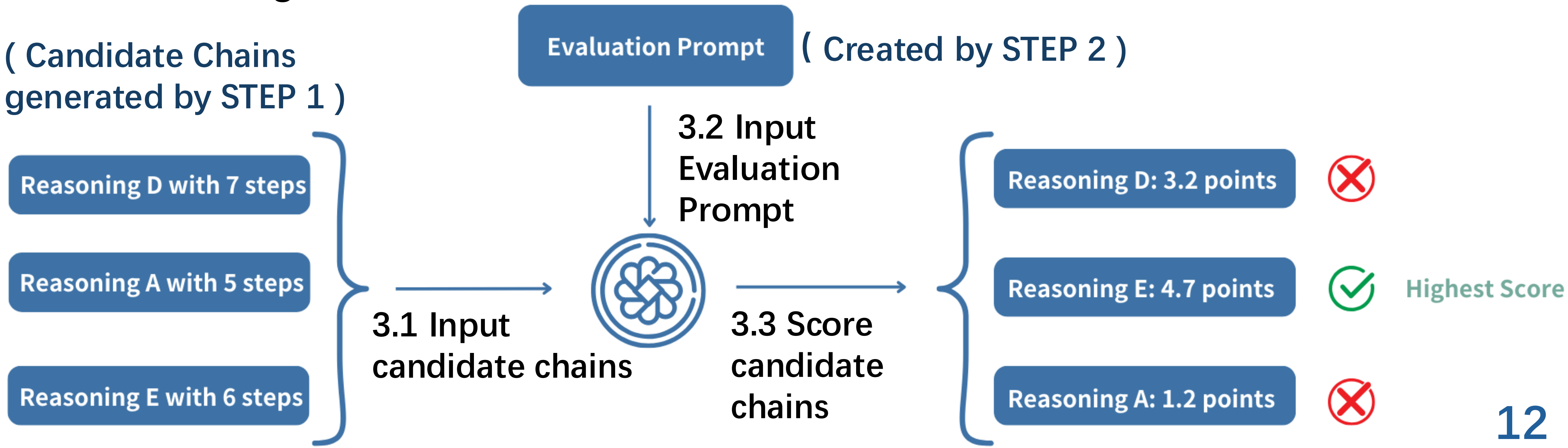
+

**Few-Shot Examples:**
Q: ...
A: ...

**Eval:**
**Step 1:**
**-Logic: ...**
**-Calculation: ...**
**Score:** 5 points **(logic:** 3points **+ Calculation:** 2 points**)**
**Step 2: ...**
**Step 3: ...**
**...**
**Average Score per Step:** 1.25 points

# Proposed Method

- **STEP 3: Evaluating and Selecting Chains**
  1. **Step-by-Step Evaluation:** Evaluate each reasoning chain individually based on both **logical coherence** and **calculation accuracy.**
  2. **Scoring Mechanism:** Assign scores using predefined **grading rules.**
  3. **Final Selection:** Compute **average score per step** and select the reasoning chain with the **highest score** as the final answer.



# Proposed Method

- **Key Advantages of Our Method:**
  - **Fine-Grained Error Evaluation:** Separates **logical errors** and calculation **errors**, ensuring detailed evaluation.
  - **Transparent & Interpretable Scoring:** Each step is **explicitly graded**, making reasoning quality **clearly measurable**.

# Table of Contents


1. Introduction
2. Previous Methods
3. Proposed Method
4. Experiments
5. Conclusions

# Experiments

- **Experiment Setup:**
  - **Dataset:** GSM8K
    - A benchmark dataset containing 8,000 **high-quality** grade-school-level **math problems**.
    - The dataset is divided into **training and test** splits, with the test set containing 1,319 problems.
  - **Model:** GPT-4o-mini
    - Chosen for its balance between **computational efficiency** and **reasoning performance**.
  - **Temperature:**
    - **Sample CoTs:** use **0.7** to ensure diversity.
    - **Evaluate CoTs:** use **0.0** to ensure deterministic evaluations.

# Experiments: Effect of $K$ on Performance

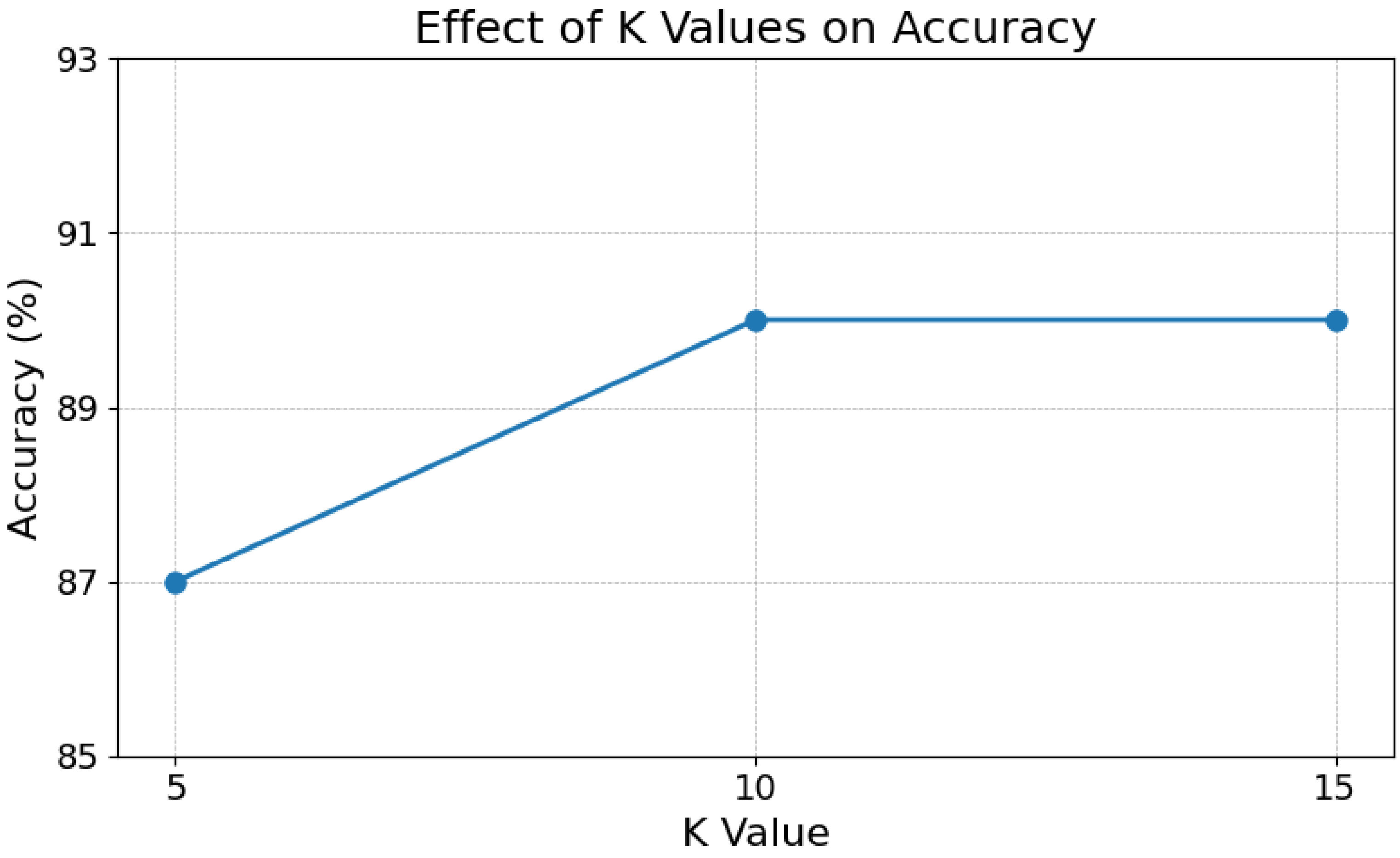
- **Purpose:** Determine the optimal **number of candidate reasoning chains ($K$)** to maximize accuracy in the proposed method.

- **Importance:** The value of $K$ directly impacts the **diversity** of reasoning paths considered and the **computational cost** of evaluating these paths.
  - **Lower $K$ :** overlook potentially valuable reasoning chains.
  - **Higher $K$ :** introduce redundancy & increase computational overhead without improving accuracy.

- **Setup:**
  - **Dataset:** 100 randomly selected instances from the GSM8K test set.
  - **Number of Samples:** 20
  - **Different Values of $K$:** 5, 10 and 15

# Experiments: Effect of $K$ on Performance

- **Result:**




- $K = 10$ provides a balanced trade-off between **accuracy** and **computational efficiency**

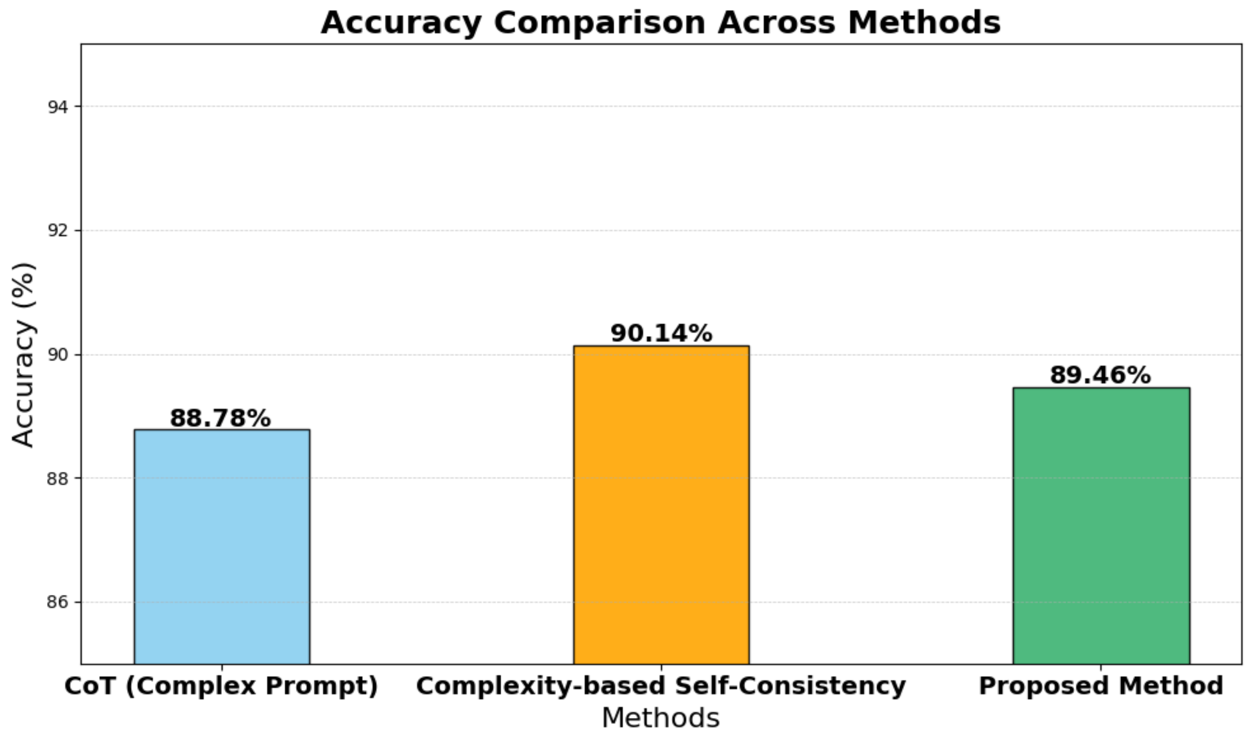
- By choosing $K = 10$, it achieves **optimal accuracy** with **minimal computational cost**, making it the recommended setting for subsequent experiments.

# Experiments: Comparison with Baselines

- **Baselines:**
  - **Chain-of-Thought Prompting (CoT):**
    - Uses a complex prompt to generate step-by-step reasoning chains
    - Relies solely on the model's output **without additional evaluation**
    - A standard benchmark for multi-step reasoning.
  - **Complexity-Based Self-Consistency:**
    - Generates 20 reasoning chains per question using a complex prompt
    - Selects the top 10 most complex chains
    - Determines the final answer via **majority voting**.
- $\boldsymbol{K}$: 10 (Based on the conclusions drawn from the last experiment)

# Experiments: Comparison with Baselines

- **Result:**




- **Higher Accuracy than CoT**: Demonstrate the effectiveness of complexity-driven candidate selection and detailed step-by-step evaluation.

- **Competitive with Complexity-Based Self-Consistency**: While slightly lower in accuracy, the proposed method offers **greater interpretability** and **fine-grained error detection**.

- **Enhanced Explainability**:
  - Complexity-Based Self-Consistency: Lacks transparency in **how the final answer is determined.**
  - Proposed method: Scoring mechanism offers **interpretable feedback**, making it easier to **identify errors** and understand why a particular reasoning chain is selected.

# Table of Contents


1. Introduction
2. Previous Methods
3. Proposed Method
4. Experiments
5. Conclusions

# Conclusions

- **Conclusion:**
  - Proposed a method integrating **complexity-driven candidate selection** with a **step-by-step evaluation,** enhancing **interpretability** and **fine-grained error detection.**
  - Designed a scoring prompt to assess **logical coherence** and **calculation accuracy**, providing detailed insights into intermediate reasoning steps.
- **Future work:**
  - Refine scoring criteria for better error detection
  - Refine few-shot examples to cover a broader range of reasoning patterns.

# References


1. Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H.Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In Advances in Neural Information Processing Systems 35: Annual Conference on Neural Information Processing Systems 2022, NeurIPS, 2022.
2. Yao Fu, Hao Peng, Ashish Sabharwal, Peter Clark, and Tushar Khot. Complexity-based prompting for multi-step reasoning. In The Eleventh International Conference on Learning Representations, ICLR, 2022.
3. Yixuan Weng, Minjun Zhu, Fei Xia, Bin Li, Shizhu He, Shengping Liu, Bin Sun, Kang Liu, and Jun Zhao. Large language models are better reasoners with self-verification. In The 2023 Conference on Empirical Methods in Natural Language Processing, EMNLP, pages 2550–2575, 2023.

# Appendix: Example Output of Complexity-Based

**Question:** If there are 3 cars in the parking lot and 2 more cars arrive, how many cars are in the parking lot?

**A:**

Let's think step by step

There are originally 3 cars.

There are 2 more cars leave, So parking lot now has 3-2 = 1 cars.

The answer is 1.

# Appendix: Example Output of Proposed Method

**Question:** If there are 3 cars in the parking lot and 2 more cars arrive, how many cars are in the parking lot?
**A:**
Let's think step by step
There are originally 3 cars.
There are 2 more cars leave, So parking lot now has 3-2 = 1 cars.
The answer is 1.

**Eval:**
**Let's evaluate step by step**

**Step 1: "There are originally 3 cars."**
**- Logic: The statement accurately reflects the initial condition given in the problem, which states there are 3 cars in the parking lot.**
**Score: 5 points (Logic: 3 points, Calculation: 2 points)**

**Step 2: "There are 2 more cars leave, So parking lot now has 3-2 = 1 cars."**
**- Logic: This step contains a crucial error. The problem states that 2 more cars arrive, not leave, which should result in addition, not subtraction. The logic does not follow from the problem statement.**
**- Calculation: The calculation is based on incorrect logic. The correct calculation, based on the problem's information about cars arriving, would be 3 + 2 = 5 cars.**
**Score: 0 points (Logic: 0 points, Calculation: 0 points)**

**Total Score: 5 points**
**Average Score per Step: 2.5**